

Aditya Kulkarni

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PROFESSIONAL SUMMARY

Software engineer specializing in LLM integration and AI evaluation infrastructure -- built a multi-model LLM platform (Claude, GPT-4o, Gemini + MCP) in production at Isomer AI and an AI quality evaluation framework benchmarking all three providers across hallucination rate, latency, and statistical significance. Daily Claude Code user; 394 commits and 7 production deployments shipping developer-facing tooling with clean REST APIs, Temporal.io agentic workflows, and CI/CD pipelines. Python and TypeScript across the full stack.

CORE COMPETENCIES

LLM APIs + MCP Integration

AI Evaluation Frameworks

Prompt Engineering

Python + FastAPI

TypeScript + Node.js

Agentic Workflows + Temporal

AWS + GCP + Docker

CI/CD + Developer Tooling

WORK EXPERIENCE

Isomer AI May 2025 - Present

Product Engineer Intern - Raleigh, NC

- Integrated **Claude, GPT-4o, and Gemini** (8+ models, 3 cloud AI providers) via MCP into a production LLM platform; implemented Temporal.io agentic workflow orchestration and real-time event pipelines enabling AI agents to integrate directly into user workflows
- Shipped 10+ developer-facing backend features across **394 commits and 7 production deployments** -- TypeScript, Node.js, Express, PostgreSQL, Redis, Auth0 (RBAC), Docker, GitHub Actions CI/CD, Sentry
- Built composable REST APIs with RFC 9457 error responses and prompt-driven document processing pipelines (PDF generation, Microsoft Graph API, S3) reviewed and extended by non-engineer collaborators

NC State University Apr 2025 - Present

Graduate Research Assistant - Raleigh, NC

- Deployed a GCP Cloud Run platform enabling 100+ researchers to explore 500GB genomic datasets; applied hyperparameter tuning and time-series ML analysis, improving model accuracy by 12%

Mphasis (HP Company) Jun 2024 - Aug 2024

Software Developer Intern

- Built 5+ AI-driven workflows including a health claim automation tool (OCR + NLP + ML); improved accuracy by 25%; automated AWS and GCP deployments with Grafana monitoring dashboards

Capgemini Jun 2023 - Aug 2023

AI/ML Intern

- Implemented distributed data pipelines (Hadoop + Spark + Kafka) reducing latency by 20%; led full SDLC for a data-driven support ticket analysis system

PROJECTS

LLM Eval Harness -- AI Quality Evaluation Framework Claude + GPT-4o + Gemini github.com/Aditya-k24/llm-eval-harness

Vendor-neutral AI quality evaluation framework benchmarking Claude Sonnet 4.6, GPT-4o, and Gemini 2.5 Pro across 3 task types (grounded QA, multi-hop QA, FEVER claim verification) on a 2,000-example frozen test set with 3 repeated runs per model. Metrics: ExactMatch, TokenF1, abstention accuracy, grounded hallucination rate, p50/p95 latency, cost/call; statistical significance via paired bootstrap CIs, McNemar test, and Holm correction. Structured prompt templates and evaluation infrastructure reusable across model versions.

Python, Anthropic/OpenAI/Google GenAI SDKs, DuckDB, Parquet, Streamlit, Docker

AlgoPulse -- Agentic AI Orchestration Platform

Temporal + Kafka + TypeScript

github.com/Aditya-k24/AlgoPulse

Event-driven agentic AI platform where AI agents integrate into async human workflows -- Temporal.io and Kafka for state synchronization and fault tolerance at 2,000 inference requests/second. Identified API-polling bottleneck mid-build and pivoted to Supabase Edge Functions for streaming AI reasoning. TypeScript, Node.js, React.

TypeScript, Node.js, React, Supabase Edge Functions, Temporal.io, Kafka

VectorLift -- Semantic Search Engine

Python + Kafka + FastAPI

github.com/Aditya-k24/VectorLift

Production-grade semantic search on MS MARCO: Kafka-driven indexing pipeline for corpus ingestion into Elasticsearch and Qdrant; FastAPI search server with per-stage latency tracking; fine-tuned sentence-transformers with in-batch negatives; 6 pipeline configurations evaluated with NDCG@10, MRR@10, MAP. Prometheus + Grafana observability throughout.

Python, FastAPI, Kafka, Elasticsearch, Qdrant, PyTorch, Sentence-Transformers, Prometheus, Grafana, Docker

Glance AI -- Compliance-Ready RAG System

RAG + Python

github.com/Aditya-k24/GlanceAI

RAG system for regulated environments (fintech/healthcare) with per-tenant vector retrieval isolation preventing context leakage and LLM hallucinations; deterministic output architecture ensuring factual correctness for compliance-sensitive AI features.

Python, LLMs, Vector Databases, RAG, per-tenant isolation

EDUCATION

North Carolina State University -- MS Computer Science

Aug 2024 - May 2026

Raleigh, NC | Courses: Artificial Intelligence, Operating Systems, Software Engineering, Algorithms, Data Science

Mumbai University -- BTech Computer Science

Aug 2020 - May 2024

CGPA: 9.32/10 | Courses: Data Structures & Algorithms, Machine Learning, Software Engineering, Big Data Infrastructure

SKILLS

AI/LLM: Claude API, OpenAI, Gemini, MCP, RAG, LangChain, Hugging Face, PyTorch, prompt engineering

Languages: Python, TypeScript, Java, JavaScript, SQL

Backend: FastAPI, Node.js, Express, REST APIs, Kafka, Temporal.io, PostgreSQL, Redis, Docker

Cloud + DevOps: AWS (S3, automation), GCP (Cloud Run), GitHub Actions CI/CD, Kubernetes, Terraform

Observability: Prometheus, Grafana, Sentry, OpenTelemetry